Kuan-Wen (Steven) Chen

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EDUCATION

Carnegie Mellon University, School of Computer Science — Pittsburgh, PA
Master of Software Engineering in Scalable Systems — Expected December 2027

National Cheng Kung University — Tainan, Taiwan
Bachelor of Science in Engineering Science — June 2024
Coursework Focus: Computer Science
CS Coursework GPA: 3.88/4.00
Selected Coursework: Data Structures, Algorithms, Operating Systems, Computer Networks, Database Systems, Computer Organization and Assembly Language, Image Recognition AI and Robotics Lab

SKILLS

Programming Languages: Python, SQL, C/C++
Backend Development: Django, REST APIs, API Design
Cloud & Developer Tools: AWS (CDK, Lambda, S3, Kinesis, Athena, Glue, CloudWatch), Docker, Git, Linux, CI/CD
ML & Infrastructure: PyTorch, AWS Neuron SDK, AWS Inferentia, Infrastructure as Code(IaC), Inference Optimization

PROFESSIONAL EXPERIENCE

Tomofun Co., Ltd. (Furbo) — Taipei, Taiwan
AI Engineer (Machine Learning Operations) Intern — July 2025–March 2026
- Refactored a legacy C-based YOLOv4 pipeline in PyTorch and compiled models with AWS Neuron SDK for deployment on AWS Inferentia2 accelerators
- Designed a hybrid inference runtime that executes tensor computations on NeuronCores and dynamic post-processing on CPUs to optimize end-to-end latency
- Built an AI inference infrastructure pipeline using AWS CDK and AWS Kinesis Data Streams, decoupling low-latency monitoring from high-throughput analytics to improve reliability; created and maintained the pipeline with IaC.
- Reduced daily stored data volume from approximately 180 GB to 20 GB using Firehose, Parquet serialization, and dynamic partitioning; enabled SQL analytics with AWS Glue and Athena
- Automated production monitoring and weekly model performance reports using CloudWatch, Lambda, SNS, and SES to support anomaly investigation and retraining decisions

GliaCloud Co., Ltd. — Taipei, Taiwan
Data Scientist Intern — January 2024–September 2024
- Built Django ETL pipelines for Google Ad Manager data and supported Metabase dashboards for account managers to monitor advertising performance and revenue metrics.
- Developed an exponential trend model to forecast monthly platform deductions from traffic and quality signals, supporting month-end revenue forecasting
- Developed statistical anomaly detection using daily, weekly, and monthly traffic patterns and integrated Slack alerts to support incident response
- Built video click-through-rate prediction models using Twelve Labs embeddings and developed a creative assistant with LangChain and LLMs to generate video concepts

PROJECTS

Autonomous Edge AI Navigation | NVIDIA Jetson Nano — September 2023–January 2024
- Deployed ResNet-50 with TensorRT FP16 on Jetson Nano, collaborating with teammates to balance inference speed and accuracy under limited compute resources
- Integrated Hough-transform-based preprocessing and background removal to reduce visual noise and support reliable road following
- Designed a steering system with vision outputs with teammates from mechanical engineering major

ECO Web Platform — February 2024–June 2024
- Led backend design in a four-person team, developing Django services, REST APIs, and database models, and the project earned the highest score in the undergraduate Database System Contest.

Badminton Serving Robot — September 2022–June 2023
- Collaborated with teammates from different engineering majors to integrate mechanical, electrical, and control software components, taking responsibility for power management and motor control
- Implemented RPM feedback and PID control with C and Arduino to stabilize launch wheel speed and improve shot consistency; the project won First Place in the 2023 NCKU Engineering Science Senior Project Contest